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CONFIDENTIAL

PHI COMMERCE PRIVATE LIMITED



GATEWAY INTERFACE SPECIFICATION DOCUMENT

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Publication Improvements

PhiCommerce invites constructive comments on the contents of this document. Please send your comments to support@phicommerce.com.

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Introduction

This document provides detailed integration process for Merchants to process payments through PayPhi Payment Gateway. This integration will allow the consumers to select payment options, provide payment instrument credentials and process payments on the Merchant website.

Direct Integration mode

Step 1: All the payment details would be collected on the Merchant website for payment transaction.

Step 2: Merchants submits the details to PayPhi sale url in a server to server call (initiateSale API call). PayPhi initiateSale API responds back with response parameters as part of API Response. Browser is not involved in this step.

Step 3: Merchant handles API response and does the following

1) Redirection (for 3DSecure on Card Issuer's domain)

When showOTPCapturePage is 'N' in the API Response

Further Steps:

1. Redirect to the redirect URI available in the initiateSale API Response with relevant parameters
2. Once Authentication and Authorization is done the response will be posted back (via browser POST) to the merchant's return URL.

2) Seamless

When showOTPCapturePage is 'Y' in the API Response

Further Steps:

1. Merchant submits generateOTP GET request to the generateOTP URI available in the initiateSale Response with relevant parameters. Response will be provided as json.
2. After successful generateOTP, merchant captures OTP on their domain and submits verifyOTP POST request to the verifyOTP URI available in the initiateSaleResponse with json request parameters.
3. Merchant can resend generateOTP request upto 3 times and verify OTP.

4. After successful verifyOTP, merchant submits authorize POST request to the authorize URI available in the initiateSale Response with json request parameters. Response will be provided as json.

Initiate Sale

The sale request is a json request.

url - pg/api/v2/initiateSale

Initiate Sale Request Parameters

Parameter	Description	Type	Size	Mandatory	
merchantId	Merchant account ID	Numeric	12	Yes	
merchantTxnNo	Txn ref Number	Alphanumeric	20	Yes	Unique ref no to identify the request. Should have only alphanumeric characters. No other characters allowed.
amount	Transaction amount	Numeric	9,2	No	9 significant and 2 decimal/precision e.g., nnnnnnnnnn.nn
currencyCode	Transaction Currency	Numeric	3	Yes	356
payType	Possible values are 0, 1 Where 0 = Standard 1 = Direct	Numeric		Yes	Currently implemented only for '1' Will give validation error if '0' provided.
customerEmailID	Email ID of customer	Alphanumeric	48	Yes	In case Customer email ID is not available, send guest@phicommerce.com
transactionType	Type of Transaction • SALE	Alphanumeric		Yes	

	• PREAUTH				
paymentMode	Payment instrument restriction list CSV, - CARD - NB - WALLET - UPI - AADHAAR - GOOGLEPAY - NACH	Alphanumeric		No	Can be used to restrict the payment options. For payType = 1, this field should indicate the exact paymentMode For AADHAAR the payType has to be 1
paymentOptions	Available options under a select payment mode	Alphanumeric			Future use
returnURL	Return URL after transaction completion	Alphanumeric	64		Call back url on which the response will be posted on payment success/rejection. If available in request, the same would be used for response else the preconfigured URL of the merchant would be used. In case the transaction involves OTP capture on merchant page then the return URL is not used.
txnDate	Transaction Date	Numeric		Yes	YYYYMMDDHHMISS
cardNo	Card Number / token Number (in case of tokenized card)	Numeric	19	C	M, If payType is '1' and paymentMode is CARD
cardExpiry	Card expiration (YYYYMM)	Numeric		C	M, If payType is '1' and paymentMode is CARD
nameOnCard	Cardholder Name	Alphanumeric	45	C	M, If payType is '1' and paymentMode is CARD
cvv	Card verification value	Numeric	4	C	M, If payType is '1' and paymentMode is CARD

					Optional in case token number is passed in cardNo field
tokenCryptogram	Cryptogram / TAVV received from TRG/TR for transacting using token number	Alphanumeric	128	C	Can contain Alphanumeric & Base64 characters.
tokenRefId	Token ReferenceID corresponding to the token number	Alphanumeric	64	O	Optional.
saveCardIndicator	Customer authorization to tokenize the card	Alpha	1	O	Y / N. Y – Indicates that the customer has opted to tokenize the card. Applicable only if the cardNo passed is an actual card no (Not applicable for token card No)
authOnlyIndicator	Indicates the transaction as an authentication only transaction.	Alpha	1	O	Y / N. Default is N. Y – The transaction is only for authentication. Save card indicator would also needs to be sent as Y for authOnly to work. This field is for supporting Rupay tokenization with Authentication (no authorization). Note: amount field is not required for authOnly transaction.
tokenizeOnAuth	Indicates whether to tokenize the card during Authentication Only transaction	Alpha	1	O	Y / N. Default is N. Y – Tokenize the card during the transaction.

					Auth only indicator would also need to be sent as Y for this to work. This field is for supporting Rupay tokenization with Authentication (no authorization).
panReferenceId	PAN reference Id of the card generated during the earlier tokenization call	Alpha	64	C	This will be used for authentication & tokenization. Either card number or panReferenceId should be present in the request. Auth Only indicator would also need to be Y for this to work.
customerUPIAlias	Customer's Alias	Alphanumeric	45	C	If paymentMode is UPI then this field will be used to pass the payer's UPI Alias (VPA)
aggregatorID	Merchant Aggregator ID	Alphanumeric		C	Only applicable for Aggregator's
customerName	Customer Name	Alphanumeric	45	No	
customerMobileNo	Customer Mobile No	Numeric	15	No	
invoiceNo	Merchant Invoice No	Alphanumeric	32	No	
orderDesc	Short description of the purchased items	Alphanumeric	45	No	For future use
addlParam1	Additional info 1	Alphanumeric	45	No	
addlParam2	Additional info 2	Alphanumeric	45	No	
addline1	Address line 1	Alphanumeric	45	No	Future use
addline2	Address line 2	Alphanumeric	45	No	Future use

cityname	City name	Alphanumeric	32	No	Future use
state	State code	Alphanumeric	32	No	Future use
zipCode	Pin code of the location	Numeric	12	No	Future use
country	Two digit country code	Alphanumeric	2	No	e.g., IN, US
txnChannel	1) WEB 2) MOBILE 3) IVR 4) POS	Alphanumeric		No	If nothing is sent then WEB is taken as default
terminalID	Terminal ID provided by acquirer	Alphanumeric	8	C	Contact support on how to use this field.
settlementProdCode		Alphanumeric	12	No	Future use
customerID	ID of customer as identified by Merchant	Alphanumeric	48	No	For Broker integration: customerID as registered in merchant. For Nach debit request this field is mandatory and should match with the customerID sent in registration of nach request.
accountNo		Alphanumeric	16	No	For Broker integration: If merchant needs to validate the account from which payment is made. If not sent, then the registered account details of customerID would be used.
accountIFSC		Alphanumeric	12	No	For Broker integration: The IFSC Code of the accountNo If not sent, then the registered account details of customerID would be used.

gstIn		Alphanumeric	15	C	GSTIN of the merchant. Required if gst details are passed by the merchant.
gst		Numeric	9,2	C	The GST amount calculated by the merchant
cgst		Numeric	9,2	C	The CGST amount
sgst		Numeric	9,2	C	The SGST amount
igst		Numeric	9,2	C	The IGST amount
cess		Numeric	9,2	C	CESS amount
gstIncentive		Numeric	9,2	C	GST Incentive
gstPercentage		Numeric	2	C	GST Percentage used.
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic
Following fields are aadhaar specific and mandatory for paymentMode of AADHAAR					
rdsId	Unique ID of the Registered device service		16		Provided by the registered device service
rdsVer	Registered device service version		12		
dpId	Unique code assigned to RD provider		24		Provided by the registered device (RD)
mi	Model of the registered device		12		-do-
mc	public key certificate of the registered device		2048		-do-
dc	Unique device code of the registered biometric device		48		-do-
paymentInstID	Aadhaar Number		12		
bankiin	IIN of the bank		6		
hmac	HMAC for the pidData		128		Provided by the RD
pidFormatType	X		1		Currently only "X" (XML) is supported
pidData	Encrypted pid data		3000		Provided by the registered device service
imageType	The fingerprint image type		3		Currently only "FMR" is supported
sessionKey	Session key used for pidData encryption		512		Provided by the RD

aadhaarCertificateID	The certificateID of the UIDAI public certificate used to encrypt the pidData		8		YYYYMMDD
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Initiate Sale Response Parameters

Will be available as json response

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure	Alphanumeric	4	M	R1000 – Request Initiated successfully Any other value indicates failure
responseDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantId	Merchant account ID	Numeric		C	Echo of request field
aggregatorID	Merchant aggregator ID	Numeric		C	Echo of request field (if available in request)
merchantTxnNo	Merchant Txn ref Number	AlphaNumeric		C	Echo of request field
redirectURI	Redirection URL		128	C	Will be available in case merchant is required to redirect to complete payment authorization. Either this field or showOTPCapturePage field will be available in response.
generateOTPURI	generateOTPurl			C	Will be available when showOTPCapturePage is 'Y'
verifyOTPURI	verifyOTPURL			C	Will be available when showOTPCapturePage is 'Y'
authorizeURI	authorizeURL			C	Will be available when

					showOTPCapturePage is 'Y'
showOTPCapturePage	Show OTP Page indicator	Char	1	C	Y / N Y – Merchant should display a page to ask for OTP. Please do contact integration team for OTP capture page requirements
tranCtx	The transaction context generated by the PG	AlphaNumeric	512	C	Needs to be sent as is to the redirectURI / authorizeURI endpoint
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Generate OTP

Merchant needs to send request if showOTPCapturePage is 'Y' in initiate sale response

The generateOTP is a GET request.

<generateOTP URI available in initiateSale Response>?tranCtx=<tranCtx available in sale Response>

Generate OTP Response Parameters

Will be available as json response

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure	Alphanumeric	4	M	0000 – Success
responseDescription	Description	Alphanumeric	45	M	
status	Status of transaction	Alphanumeric		M	

merchantId	Merchant account ID	Numeric		C	merchantId of original transaction
aggregatorID	Merchant aggregator ID	Numeric		C	aggregatorId of original transaction
merchantTxnNo	Merchant Txn ref Number	AlphaNumeric		C	merchantTxnNo of original transaction
secureHash	Hash value	Alphanumeric	64	C	Check Hash Calculation in Appendix

Verify OTP

Merchant needs to send request after generate OTP request is successful

The verifyOTP request is a POST request.

<verifyOTP URI available in initiateSale Response>

Json Request Message as below:

Verify OTP Request Parameters

Will be sent as json request

Parameter	Description	Type	Size	Mandatory	
tranCtx	The transaction context generated by the PG	AlphaNumeric	512	Yes	tranCtx sent in initiateSale Response
otp	OTP received by customer	Numeric	6	Yes	Merchant needs to provide OTP received by customer

Verify OTP Response Parameters

Will be available as json response

Parameter	Description	Type	Size	Mandatory	
-----------	-------------	------	------	-----------	--

responseCode	Response code indicating success/failure	Alphanumeric	4	M	0000 – Success
responseDescription	Description	Alphanumeric	45	M	
status	Status of transaction	Alphanumeric		M	
merchantId	Merchant account ID	Numeric		C	merchantId of original transaction
aggregatorID	Merchant aggregator ID	Numeric		C	aggregatorId of original transaction
merchantTxnNo	Merchant Txn ref Number	AlphaNumeric		C	merchantTxnNo of original transaction
nwAuthRefNo	Network Auth ref No	Char	64	C	Currently only applicable for rupay cards. This will be available if customer has opted for saveCard & authOnly as Y. This value needs to be used when initiating tokenization with RuPay's Token Server
panReferenceId	Pan Reference ID of the card number for this token is generated	Alpha	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
maskedCardNo	Masked Card number	Alpha	19	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
tokenReferenceId	Token reference Id to identify a token for a card number	Alphanumeric	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
tokenReferenceIdHash	Token reference id hash to validate the token reference id	Alphanumeric	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
cardExpiry	Card expiration (YYYYMM)	Numeric		C	It will be present in the response if

					tokenizeOnAuth was Y in the initiateSale request
secureHash	Hash value	Alphanumeric	64	C	Check Hash Calculation in Appendix

Authorize

Merchant needs to send request after verify OTP request is successful

The authorize request is a POST request.

<authorizeURI available in initiateSale Response>

Authorize Request Parameters

Will be sent as json request

Parameter	Description	Type	Size	Mandator y	
tranCtx	The transaction context generated by the PG	AlphaNum eric	512	Yes	tranCtx sent in initiateSale Response

Authorize Response Parameters

Will be available as json response

Parameter	Description	Type	Siz e	Mandator y	
responseCode	Response code indicating success/failure	Alphanumeri c	4	M	000 and 0000 – Success R1000 – Request Initiated successfully (mostly used for UPI Initiation, NACH where the

					payment is out of band.) Any other value indicates failure
respDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantId	Merchant account ID	Numeric		C	Echo of request field
aggregatorID	Merchant aggregator ID	Numeric		C	Echo of request field (if available in request)
merchantTxnNo	Merchant Txn ref Number	Alphanumeric		C	Echo of request field
txnID	The unique ref number generated by the PG	Numeric	24	C	
paymentDateTime	Authorization date time	Numeric		C	YYYYMMDDHHMISS
paymentID	The authorization/payment ID generated by the authorization system	Numeric	24	C	
paymentMode	The payment mode used CARD, NB, UPI, etc.,	Alphanumeric	12	C	Available only for merchants who have opted for this. Determined using configuration or using the sale request url used.
paymentSubInstType	The payment sub instrument type. Will contain the bank name in case of payment using NB. In case of card payment this field will be either DC or CC	Alphanumeric	64	C	-do-
amount	The amount paid	Numeric	9,2	C	-do-
txnChannel	The channel used for transaction (POS, WEB)	Alphanumeric	5	C	For POS, this will be returned always for successful txns.

					If not available it is to be treated as WEB transaction.
customerMobileNo	The customer mobile number as sent by merchant	Alphanumeric	48	C	Echo of request field. Available based on configuration / sale url used
customerEmailID	The emailID of the customer as sent by merchant	Alphanumeric	48	C	Echo of request field Available based on configuration / sale url used
addlParam1	Additional info 1	Alphanumeric		C	Echo of request field
addlParam2	Additional info 2	Alphanumeric		C	Echo of request field
convenienceFee	Convenience fee charged to customer	Numeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
serviceTax	Service tax	Numeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
cardNetwork	Card Network	Alphanumeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
nwAuthRefNo	Network Auth ref No	Char	64	C	Currently only applicable for rupay cards. This will be available if customer has opted for saveCard. This value needs to be

					used when initiating tokenization with RuPay's Token Server
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Authorization Redirect Request

Merchant needs to redirect to the redirect URI in the following request format when showOTPCapturePage is 'N' is sale response

<redirectURL available in Sale Response>?tranCtx=<tranCtx available in sale Response>

The payment response would be posted back to the merchants return URL (in form url encoded format) as form post.

Payment Response Parameters

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure	Alphanumeric	4	M	000 and 0000 – Success R1000 – Request Initiated successfully (mostly used for UPI Initiation, NACH where the payment is out of band.)

					Any other value indicates failure
respDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantId	Merchant account ID	Numeric		C	Echo of request field
aggregatorID	Merchant aggregator ID	Numeric		C	Echo of request field (if available in request)
merchantTxnNo	Merchant Txn ref Number	Alphanumeric		C	Echo of request field
txnID	The unique ref number generated by the PG	Numeric	24	C	
paymentDateTime	Authorization date time	Numeric		C	YYYYMMDDHHMIS S
paymentID	The authorization/payment ID generated by the authorization system	Numeric	24	C	
paymentMode	The payment mode used CARD, NB, UPI, etc.,	Alphanumeric	12	C	Available only for merchants who have opted for this. Determined using configuration or using the sale request url used.
paymentSubInstType	The payment sub instrument type. Will contain the bank name in case of payment using NB. In case of card payment this field will be either DC or CC	Alphanumeric	64	C	-do-
amount	The amount paid	Numeric	9,2	C	-do-
txnChannel	The channel used for transaction (POS, WEB)	Alphanumeric	5	C	For POS, this will be returned always for successful txns. If not available it is to be treated as WEB transaction.

customerMobileNo	The customer mobile number as sent by merchant	Alphanumeric	48	C	Echo of request field. Available based on configuration / sale url used
customerEmailID	The emailID of the customer as sent by merchant	Alphanumeric	48	C	Echo of request field Available based on configuration / sale url used
addlParam1	Additional info 1	Alphanumeric		C	Echo of request field
addlParam2	Additional info 2	Alphanumeric		C	Echo of request field
convenienceFee	Convenience fee charged to customer	Numeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
serviceTax	Service tax	Numeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
cardNetwork	Card Network	Alphanumeric		C	Applicable only if merchant has specified for this additional field in the returnUrl addlResParams parameter
nwAuthRefNo	Network Auth ref No	Char	64	C	Currently only applicable for rupay cards. This will be available if customer has opted for saveCard. This value needs to be used when initiating tokenization with

					RuPay's Token Server
panReferenceId	Pan Reference ID of the card number for this token is generated	Alpha	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
maskedCardNo	Masked Card number	Alpha	19	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
tokenReferenceId	Token reference Id to identify a token for a card number	Alphanumeric	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
tokenReferenceIdHash	Token reference id hash to validate the token reference id	Alphanumeric	64	C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
cardExpiry	Card expiration (YYYYMM)	Numeric		C	It will be present in the response if tokenizeOnAuth was Y in the initiateSale request
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Note: Only POST parameters are considered for secureHash calculation. Query parameters are not considered for secureHash calculation.

Sample Response form

```
<form id='myForm' name='myForm' action='${merchantReturnURL}' method='post'
enctype="application/x-www-form-urlencoded">
```

```
<input type="hidden" name="responseCode" value="000" />
```

```
<input type="hidden" name="respDescription" value="SUCCESS" />
```

```
<input type="hidden" name="merchantId" value="M000001" />
```

```
<input type="hidden" name="merchantTxnNo" value="M0099999221" />
<input type="hidden" name="txnID" value="T1472640294491" />
<input type="hidden" name="paymentDateTime" value="20160831041454" />
<input type="hidden" name="paymentID" value="006503" />
<input type="hidden" name="secureHash"
value="39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2" />
</form>
```

Payment Advice

In case merchant wants payment status to be updated to its system, Phi Transaction Gateway can push the status of the payment. Payment advice is same as payment response but only used by the payment gateway to inform the merchant system of a later change in transaction status which can occur after the online transaction has completed. This will be a server to server API call (not involving the browser). Merchant has to expose a URL for consuming the payment advice. The parameters will be sent as POST request and the parameters will be same as that used for the online payment response.

Any response with http status code of 200 is considered as successful delivery of the advice. Advice can be configured to be retried in case of unsuccessful delivery (non 200 http status code).

Note: The addlResParams if specified in returnUrl are only used for redirection parameters and not used for payment advice.

Payment advice message by default would be posted to the advice url of the merchant with content-type of "application/x-www-form-urlencoded". In this format the request body would contain all the parameters in name value pairs. The merchant server can handle this in the same manner how it handles any form post.

```
responseCode=0000&respDescription=Transaction%20successful&merchantId=T_05022&merchantTxnNo=638948982&txnID=T002473066701&paymentDateTime=20210817060032&paymentID=700788003737&paymentSubInstType=CC&paymentMode=Card&amount=51.00&secureHash=40567b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2de
```

In case the merchant has opted for json format (during merchant onboarding). The same above advice would be sent in json format. For json message the content type would be "application/json"

```
{
  "responseCode": "0000",
  "respDescription": "Transaction successful",
  "merchantId": "T_05022",
  "merchantTxnNo": "638948982",
  "txnID": "T002473066701",
  "paymentDateTime": "20210817060032",
  "paymentID": "700788003737",
  "paymentSubInstType": "CC",
  "paymentMode": "Card",
  "amount": "51.00",
  "secureHash": "40567b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2de"
}
```

Any response with http status code of 200 is taken as a successful response.

Settlement Advice

Settlement advice will be a server to server API call initiated by the PayPhi system. Merchant has to expose a URL for consuming the settlement advice. The parameters will be sent as POST request (with json data in POST body).

Request Parameters

Parameter	Description	Type	Size	Mandatory	
-----------	-------------	------	------	-----------	--

aggregatorID	Merchant aggregator ID	Alphanumeric	12	C	Echo of request field (if available in original payment request)
merchantId	Merchant account ID	Alphanumeric	12	Yes	
merchantTxnNo	Merchant Txn ref Number	Alphanumeric	12	Yes	Echo of request field, the merchant txn reference no sent by the merchant while initiating the payment request. For payment against invoices (orders created in PhiCommerce system), this field will be an auto generated field.
invoiceNos	List of Invoices (comma separated) for which the payment was done	Alphanumeric	160	C	Echo of request field In case of payment against multiple invoices, this field will contain comma separated list of invoice reference nos.
paymentMode	The payment mode/type used for the payment	Alphanumeric	32	C	e.g., CARD, NB,

					UPI, AADHAAR, BharatQR
paymentSubInstType	The sub type for the payment mode used.	Alphanumeric	32	C	If paymentMode is card then possible values would be DC, CC N.A for other payment modes.
txnChannel	The txn channel used (Applicable only for CARD)	Alphanumeric	32	C	Possible values are POS/WEB.
txnAmount		Amount		C	Applicable only if the charges are paid by Merchant & certified to receive this field
txnCharges		Amount		C	Applicable only if the charges are paid by Merchant & certified to receive this field
serviceTax		Amount		C	Applicable only if the charges are paid by Merchant & certified to receive this field
settlementID	The settlementID / credit reference no linked to this invoice settlement	Alphanumeric	32	Yes	
settlementDate	Settlement Date	Alphanumeric	8	Yes	YYYYMMDD

settlementAccount	Settlement Account	Alphanumeric	21	Yes	Available only if the merchant has certified for receiving this field
settlementAccountIFSC	Settlement Account IFSC Code	Alphanumeric	8	Yes	Available only if the merchant has certified for receiving this field
settledAmount	Settlement Amount	Amount	8	Yes	Available only if the merchant has certified for receiving this field
cardNetwork	Card network	Alpha	8	Yes	Available only if the merchant has certified for receiving this field and if the paymentMode is Card Possible values are VISA MC AMEX RUPAY
addlParam1	Additional Param sent in initial sale request	Alphanumeric	64	C	Echo of request field (if present in request)
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Request would be in json format. Sample request is given below

```
{
  "aggregatorID": "A001",
  "merchantId": "123243",
  "merchantTxnNo": "RF12937",
  "invoiceNos" : "343223",
  "settlementID": "A00121-20171101-010",
  "settlementDate": "20171101",
  "paymentMode": "card",
  "paymentSubInstType": "DC",
  "cardNetwork": "VISA",
  "txnChannel": "WEB",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```

Any response with http status code of 200 is taken as a successful response.

Generate QR Request Parameters

Parameter	Description	Type	Size	Mandatory	
merchantId	Merchant account ID	Numeric	12	Yes	
merchantRefNo	Txn ref Number	Alphanumeric	12	Yes	Unique ref no to identify the request.
amount	Transaction amount	Numeric	9,2	Yes	9 significant and 2 decimal/precision e.g., nnnnnnnnn.nn
currency	Transaction Currency Code	Numeric	3	Yes	356
emailID	Email ID of customer	Alphanumeric	36	Yes	In case Customer email ID is not available, send quest@phicommerce.com
mobileNo	Customer Mobile No	Numeric	15		
invoiceNo	Merchant Invoice No	Alphanumeric	32		
requestType	Request Type	Alphanumeric	12		Applicable values are 1) "UPIQR" 2) "BharatQR"
aggregatorID	Merchant Aggregator ID	Alphanumeric			Only applicable for Aggregator's

taxableValue	Taxable amount	Numeric	9,2	C	
gstIn		Alphanumeric	15	C	GSTIN of the merchant. Required if gst details are passed by the merchant.
gst		Numeric	9,2	C	The GST amount calculated by the merchant
cgst		Numeric	9,2	C	The CGST amount
sgst		Numeric	9,2	C	The SGST amount
igst		Numeric	9,2	C	The IGST amount
cess		Numeric	9,2	C	CESS amount
gstIncentive		Numeric	9,2	C	GST Incentive
gstPercentage		Numeric	2	C	GST Percentage used.
customerID	Customer ID	Alphanumeric	15	O	
customerName	Customer Name	Alphanumeric	45	O	
invoiceDate	Invoice Date	Numeric	8	O	YYYYMMDD
postTillNo	The till Number associated with the terminal	AN	12	No	For future use
terminalId	The terminal ID to which the order needs to be attached	AN	8	No	For future use
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

GenerateQR Response Parameters

Parameter	Description	Type	Size	Mandatory	
returnCode	Return code indicating success/failure of the QR generation request	Numeric	3	M	200 – Success Any other value failure

respDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantID	Merchant account ID	Alphanumeric		C	Echo of request field
aggregatorID	Merchant account ID	Alphanumeric		C	Echo of request field
merchantRefNo	Merchant Txn ref Number	Alphanumeric		C	Echo of request field
bharatQR	Generated BharatQR string	ANS		C	
upiQR	Generated UPI QR String	ANS		C	

Sample response (json) of generate QR request

```
{
  "respHeader": {
    "returnCode": "200",
    "desc": "Success",
  },
  "respBody": {
    "merchantId": "123243",
    "merchantRefNo" : "343223",
    "aggregatorID": "123243",
    "bharatQR" : "dfads?asdfa=dfad",
    "upiQR": "upi?pay=aa&pn=sad&amt=10"
  }
}
```

MerchantCreditToCustomer Request Parameters

The request is to be posted as json in a server to server call from an whitelisted IP address.

Parameter	Description	Type	Size	Mandatory	
aggregatorID	Merchant Aggregator ID	Alphanumeric			Only applicable for Aggregator's

merchantId	Merchant account ID	Numeric	12	Yes	
merchantRefNo	Txn ref Number		12	Yes	Unique ref no to identify the request.
Amount	Amount that needs to be credited to the customer	Numeric	9,2	Yes	9 significant and 2 decimal/precision e.g., nnnnnnnnn.nn
Currency	Transaction Currency Code	Numeric	3	Yes	356
customerPAN	Cardnumber to which the amount is to be credited	Numeric	19	C	This field is required if credit to card is to be done
customerPANExpiry	Card expiry date		6	C	YYYYMM This field is required if credit to card is to be done
firstName	First Name of customer	Alphanumeric	32	C	This field is required if credit to card is to be done
lastName	Last name of customer	Alphanumeric	32	C	This field is required if credit to card is to be done
addressLine1	Address Line1	Alphanumeric	128	C	This field is required if credit to card is to be done
City	City of residence	Alphanumeric	48	C	This field is required if credit to card is to be done
postalCode	Postal code	Alphanumeric	6	C	This field is required if credit to card is to be done
emailID	Email ID of customer	Alphanumeric	36	Yes	In case Customer email ID is not available, send quest@phicommerce.com
mobileNo	Customer Mobile No	Numeric	15	Yes	10 digit mobile phone number
beneficiaryIFSC	Customer bank IFSC Code	Alphanumeric	11	C	11 digit IFSC code of the beneficiary (customer) This field is required if credit is to be done to account using IFSC / AccNo.
beneficiaryAccNo	Customer Account No	Alphanumeric	21	C	This field is required if credit is to be done to

					account using IFSC / AccNo.
addiParam1	Additional param	Alphanumeric	32		This field will be echoed back in response.
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

MerchantCreditToCustomer Response Parameters

Parameter	Description	Type	Size	Mandatory	
returnCode	Return code indicating success/failure of the QR generation request	Numeric	3	M	200 – Success Any other value failure
respDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantID	Merchant account ID	Alphanumeric		C	Echo of request field
aggregatorID	Merchant account ID	Alphanumeric		C	Echo of request field
merchantRefNo	Merchant Txn ref Number	Alphanumeric		C	Echo of request field
txnID	The unique ref number generated by the PG	Numeric	24	C	Transaction ID generated by phicommerce
paymentDateTime	Authorization date time	Numeric		C	YYYYMMDDHHMISS
txnAuthID	The authorization/payment ID generated by the authorization system	Numeric	24	C	AuthID generated by the bank.
addiParam1	Additional info 1	Alphanumeric		C	Echo of request field
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Sample request/response (json)

Request

```
{  
    "merchantId": "123243",  
    "merchantRefNo" : "343223",  
    "aggregatorID": "123243",  
    "amount" : "124.75",  
    "currency": "356",  
    "mobileNo": "1234567890",  
    "emailID": "guest@phicommerce.com",  
    "beneficiaryIFSC": "ICIC1234221",  
    "beneficiaryAccNo": "2323232321090",  
    "secureHash": "abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234"  
}
```

Response

[illegible]

NACH Registration Request Parameters

The request is to be posted as json in a server to server call from an whitelisted IP address.

Parameter	Description	Type	Size	Mandatory	
aggregatorID	Merchant Aggregator ID	Alphanumeric			Only applicable for Aggregator's
merchantId	Merchant account ID	Numeric	12	Yes	
customerID	Id of the customer who is registering	Alphanumeric	32	Yes	Unique no to identify customer in merchant

	the NACH mandate				system. Note: Only one active mandate can be present against a customer for the merchant.
reqType	Type of NACH Mandate request	Alpha	10	Yes	CREATE, AMEND, CANCEL Currently only CREATE supported.
customerMobileNo		Numeric	10	Yes	Mobile no without 91 prefix.
customerEmailId	EmailID of the customer		48	Yes	
customerName	Customer Name as registered in the bank of given IFSC	Alphanumeric	48	No	
customerBankIFSC	Customer bank IFSC Code	Alphanumeric	11	No	11 digit IFSC code of the customer's bank
customerBankAccNo	Customer Account No	Alphanumeric	21	No	
amount	Amount that needs to be credited to the customer	Numeric	9,2	Yes	9 significant and 2 decimal/precision e.g., nnnnnnnnn.nn
startDate	Start date of Nach debit	Numeric		Yes	YYYYMMDD Should be greater than day of raising request.
endDate	End date of Nach debit			No	If not sent, then NACH request will be valid Until Cancellation. If sent then the format should be YYYYMMDD Should be greater than startDate
frequency	Nach Debit frequency	Alpha	2	Yes	Possible values M - Monthly, SA – Semi Annual, Y - Yearly,

					Q - Quarterly, V – Variable (as and when required)
debitType	Type of debit	Alpha	10	No	Possible values FIXED, MAXAMOUNT
returnURL	The return URL		128	No	If specified, the merchant would be notified on this URL on update of registration request & on acceptance/rejection of the NACH registration by the BANK. Refer the NACH Registration Completion Response parameters The response parameters will be posted as form values via browser form post.
addlParam1	Additional info 1	Alphanumeric	45	No	
addlParam2	Additional info 2	Alphanumeric	45	No	
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

NACH Registration Response Parameters

Parameter	Description	Type	Size	Mandatory	
returnCode	Return code indicating success/failure of the request	Numeric	4	Yes	0000 – Success Any other value failure
respDescription	Description of failure	Alphanumeric	45	Yes	Ignore for success response.
merchantID	Merchant account ID	Alphanumeric		No	Echo of request field
aggregatorID	Merchant account ID	Alphanumeric		No	Echo of request field

customerID	Customer ID of the merchant	Alphanumeric		No	Echo of request field
redirectURL	The URL to redirect the customer to complete the registration request.	Alphanumeric	250	No	Applicable only if returnCode is success.
addIPParam1	Additional info 1	Alphanumeric	45	No	
addIPParam2	Additional info 2	Alphanumeric	45	No	
secureHash	Hash value	Alphanumeric	64	No	Check Hash Calculation in Appendix

Sample request/response (json)

Request

```
{  
    "merchantId": "123243",  
    "customerID" : "343223",  
    "aggregatorID": "123243",  
    "reqType": "CREATE",  
    "customerMobileNo": "1234567890",  
    "customerEmailId": "guest@phicommerce.com",  
    "customerBankIFSC": "ICIC1234221",  
    "customerBankAccNo": "2323232321090",  
    "amount" : "50000",  
    "startDate": "20200501",  
    "endDate": "20230501",  
    "frequency": "V",  
    "debitType" : "MAXAMOUNT",  
    "secureHash": "abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234"  
}
```

Response

```
{  
    "returnCode": "0000",  
    "respDescription": "Transaction Successful",  
    "merchantId": "123243",  
    "customerID": "343223",  
    "aggregatorID": "123243",  
    "redirectURL": "https://www.payph.in/dfd123",  
    "secureHash": "abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234"  
}
```

NACH Cancellation Request Parameters

The request is to be posted as json in a server to server call from an whitelisted IP address.

Parameter	Description	Type	Size	Mandatory	
aggregatorID	Merchant Aggregator ID	Alphanumeric			Only applicable for Aggregator's
merchantId	Merchant account ID	Numeric	12	Yes	
customerID	Id of the customer who is registering the NACH mandate	Alphanumeric	32	Yes	Unique no to identify customer in merchant system. Note: Only one active mandate can be present against a customer for the merchant.
reqType	Type of NACH Mandate request	Alpha	10	Yes	CANCEL
notificationURL	The URL on which the cancellation acceptance to be notified to the merchant		128	No	If specified, the merchant would be notified on this URL on acceptance/rejection of cancellation request by the BANK. Refer the NACH Registration/Cancellation Completion Response parameters The response parameters will be posted as form values via browser form post.
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

NACH Cancellation Response Parameters

Parameter	Description	Type	Size	Mandatory	
returnCode	Return code indicating acceptance of the request.	Numeric	4	Yes	0000 – Success Any other value failure Actual cancellation of mandate at bank end will happen later and will be notified to merchant using the notificationURL given in cancellation request
respDescription	Description of failure	Alphanumeric	45	Yes	Ignore for success response.
merchantId	Merchant account ID	Alphanumeric		No	Echo of request field
aggregatorID	Merchant account ID	Alphanumeric		No	Echo of request field
customerID	Customer ID of the merchant	Alphanumeric		No	Echo of request field
requestAcceptedTime	Date time at which the cancel request was accepted	Numeric	14		YYYYMMDDHHMISS
secureHash	Hash value	Alphanumeric	64	No	Check Hash Calculation in Appendix

Sample request/response (json)

Request

```
{  
    "merchantId": "123243",  
    "customerID" : "343223",  
    "aggregatorID": "123243",  
    "reqType": "CANCEL",  
    "secureHash": "abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234"  
}
```

Response

```
{  
    "returnCode": "0000",  
    "respDescription": "Request Successful",  
    "merchantId": "123243",  
    "customerID" : "343223",  
    "aggregatorID": "123243",  
    "secureHash": "abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234abcd1234"  
}
```

NACH Registration/Cancellation Completion Response Parameters

The following parameters will be posted to the merchant's page (specified in the returnUrl parameter of NACH Registration Request).

The returnURL specified in the NACH registration request will be used to notify the merchant of the various status updates of the registration request. E.g. Initially the status would be NEW on entry of information by the user. Once the NACH registration is accepted by the customer's bank, it would turn to ACCEPTED. In case of rejection, the status would be updated to REJECTED. Similarly in case the mandate is cancelled by the customer the status would be updated to CANCELLED. The merchant would be notified in each of these updates using the return URL.

Note:

The NEW status is usually updated through browser redirect. The subsequent status are updated using server to server call on the same returnUrl.

System uses POST method with content type as "application/x-www-form-urlencoded" to post the below parameters via browser form post or via S2S call.

Parameter	Description	Type	Size	Mandatory	
responseCode	Return code indicating success/failure of the request	Numeric	4	Yes	0000/R1000 – Success Any other value failure

respDescription	Description of failure	Alphanumeric	45	Yes	Ignore for success response.
merchantID	Merchant account ID	Alphanumeric		No	Echo of request field
aggregatorID	Merchant account ID	Alphanumeric		No	Echo of request field
customerID	Customer ID of the merchant	Alphanumeric		No	Echo of request field
mandateType	The mandate type chosen by the customer	Alpha	12	Yes	Possible values are auto, powered
onlineMandate	Whether online mode was opted by the customer to register the mandate with bank (instead of paper form upload)	Alpha	1	No	Possible values are Y, N Default is N. If online mandate is opted by the customer then the max amount for subsequent transactions against this mandate should be less than 10L.
status	The NACH registration status	Alpha	12	Yes	Possible values are NEW, ACCEPTED, REJECTED, CANCELLED
addlParam1	Additional info 1	Alphanumeric	45	No	
addlParam2	Additional info 2	Alphanumeric	45	No	
secureHash	Hash value	Alphanumeric	64	No	Check Hash Calculation in Appendix

Refund/Auth/Void Request Parameters

This is a server to server API Call which uses POST method with content type as "application/x-www-form-urlencoded"

Parameter	Description	Type	Size	Mandatory	
merchantID	Merchant account ID	ANS	12	Yes	
aggregatorID	Aggregator ID	ANS	12		Required if the original txn was initiated from Aggregator
merchantTxnNo	Txn ref Number	Alphanumeric	12	Yes	The new refNo for this request.
originalTxnNo	Original txn refNo	Alphanumeric			The refNo of the initial transaction for which the refund/auth/void is initiated
amount	Amount to be authorized or refunded to the customer	Numeric	9,2	Yes	9 significant and 2 decimal/precision This cannot be more than the original transaction amount This field should be set to 0 for void.
transactionType	Type of Transaction REFUND AUTH VOID	Alphanumeric		Yes	
addlParam1	Additional info 1	Alphanumeric	45		
addlParam2	Additional info 2	Alphanumeric	45		
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

Refund/Auth Response Parameters

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure	Numeric	3	M	000 – Success Any other value failure
respDescription	Description of failure	Alphanumeric	45	M	Ignore for success response.
merchantId	Merchant account ID	Alphanumeric		C	Echo of request field
merchantTxnNo	Merchant Txn ref Number	Alphanumeric		C	Echo of request field
txnID	The unique ref number generated by the PG	Numeric	24	C	
paymentDateTime	Authorization date time	Numeric		C	YYYYMMDDHHMISS
txnAuthID	The authorization/payment ID generated by the authorization system	Alphanumeric	24	C	
addlParam1	Additional info 1	Alphanumeric		C	Echo of request field
addlParam2	Additional info 2	Alphanumeric		C	Echo of request field
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Transaction Status Parameter

Parameter	Description	Type	Size	Mandatory	
merchantID	Merchant account ID	ANS	12	Yes	
aggregatorID	Aggregator ID	ANS	12		Required if the original txn was initiated from Aggregator
merchantTxnNo	Txn ref Number	Alphanumeric	12	Yes	Same as originalTxnNo.
originalTxnNo	Original Txn ref Number		12	Yes	The original refNo for which status is requested.
transactionType	Type of Transaction STATUS	Alphanumeric		Yes	
secureHash	Hash value	Alphanumeric	64	No	Check appendix for hash logic

This is a server to server API Call which uses POST method with content type as "application/x-www-form-urlencoded"

Transaction Status Response Parameters

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure of the transaction status request		3	M	000 / 0000 – Success Any other value indicates status request failure / original transaction not found For status of the original txn always check txnStatus and txnResponseCode
respDescription	Description of failure		45	M	Ignore for success response.
merchantId	Merchant account ID			C	Echo of request field
aggregatorID	Merchant Aggregator ID			C	Echo of request field
merchantTxnNo	Merchant Txn ref Number			C	Echo of request field
txnStatus	Status of original txn		3	C	REQ – Request received and in process SUC – Transaction Successful REJ – Transaction Rejected ERR – Error in transaction process
txnResponseCode	The response code for the original txn		24	C	Original txn response code 000 / 0000 – Success Any other value indicates failure
txnRespDescription	The unique ref number generated by the gateway		24	C	
txnID	The unique ref number generated by the gateway		24	C	

paymentDateTime	Authorization date time			C	YYYYMMDDHHMISS
txnAuthID	The authorization/payment ID generated by the authorization system		24	C	
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Settlement Status Request Parameter

Settlement status API should be used only to check the settlement status of the transaction and never used for checking the payment status of original transaction. The payment transaction gets reflected in settlement only after the settlement cycle which might take anywhere upto 12hrs.

Parameter	Description	Type	Size	Mandatory	
merchantID	Merchant account ID	ANS	12	Yes	
aggregatorID	Aggregator ID	ANS	12	No	
originalTxnNo	Original Txn ref Number		20	Yes	The original merchant refNo for which status is requested.
transactionType	Type of Transaction SETTLSTATUS	Alphanumeric		Yes	
secureHash	Hash value	Alphanumeric	64	No	Check appendix for hash logic

This is a server to server API Call which uses POST method with content type as "application/x-www-form-urlencoded"

Settlement Status Response Parameters

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure of the transaction status request		3	M	000 / 0000 – Success Any other value indicates request failure / transaction not found in settlement system. For settlement status of the txn always check settlementStatus
respDescription	Description of failure		45	M	Ignore for success response.
merchantId	Merchant account ID			C	Echo of request field
aggregatorID	Merchant Aggregator ID			C	Echo of request field
merchantTxnNo	Merchant Txn ref Number			C	Echo of request field
settlementStatus	Status of settlement		3	C	NSD – Not yet settled STD – Settled
txnStatus	Status of original txn		3	C	REQ – Request received and in process SUC – Transaction Successful REJ – Transaction Rejected ERR – Error in transaction process
txnResponseCode	The response code for the original txn		24	C	Original txn response code 000 / 0000 – Success Any other value indicates failure
txnID	The unique ref number generated by the gateway		24	C	

settlementID	Settlement ID/UTR No		24	C	
settlementDate	Date of settlement			C	YYYYMMDD
txnAmount	The transaction amount	Numeric		C	
txnCharges	Total charges	Numeric		C	Charges including tax
serviceTax	Service tax paid	Numeric		C	
settledAmount	Amount settled	Numeric		C	The amount settled for this transaction
settlementAccount	Account No		24	C	Account no to which the txn has been settled into
settlementAccount IFSC	IFSC Code of the account		11	C	The IFSCCode of the account No.
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Settlement Summary Request Parameter

Settlement status API should be used only to check the settlement status of the transaction and never used for checking the payment status of original transaction. The payment transaction gets reflected in settlement only after the settlement cycle which might take anywhere upto 12hrs.

Parameter	Description	Type	Size	Mandatory	
merchantID	Merchant account ID	ANS	12	No	Either of merchantID OR aggregatorID should be present
aggregatorID	Aggregator ID	ANS	12	No	-do-
settlementDate	Settlement Date	ANS	10	Yes	YYYY-MM-DD
transactionType	Type of Transaction SETTLEMENTSUMMARY	Alphanumeric		Yes	
secureHash	Hash value	Alphanumeric	64	No	Check appendix for hash logic

This is a server to server API Call which uses POST method with content type as "application/x-www-form-urlencoded"

Settlement Summary Response Parameters

Parameter	Description	Type	Size	Mandatory																																																	
error_code	Response code		3	M	000 / 0000 – Success Any other value indicates request failure / transaction not found in settlement system. For settlement status of the txn always check settlementStatus																																																
merchantId	Merchant account ID			C	Echo of request field																																																
aggregatorID	Merchant Aggregator ID			C	Echo of request field																																																
settlementDate	Settlement Date			C	Echo of request field																																																
Payouts	All the settlements performed on the settlement Date		3	C																																																	
<table> <tr> <td>merchantID</td><td>Merchant account ID</td><td></td><td></td><td></td><td>Echo of request field</td></tr> <tr> <td>account_no</td><td>Account number where the amount is settled</td><td>AlphaNumeric</td><td></td><td></td><td></td></tr> <tr> <td>ifsc_code</td><td>IFSC code of the account number</td><td>AlphaNumeric</td><td></td><td></td><td></td></tr> <tr> <td>pay_amount</td><td>Amount settled</td><td>Numeric</td><td></td><td></td><td></td></tr> <tr> <td>settlementID</td><td>Settlement id</td><td></td><td>32</td><td></td><td>Settlement ID for merchant or service provider</td></tr> <tr> <td>settlement_currency</td><td>Settlement Currency</td><td>Numeric</td><td>3</td><td></td><td>356</td></tr> <tr> <td>trans_currency</td><td>Settlement Currency</td><td>Numeric</td><td>3</td><td></td><td>356</td></tr> <tr> <td>utr_no</td><td>UTR number</td><td></td><td>24</td><td></td><td></td></tr> </table>						merchantID	Merchant account ID				Echo of request field	account_no	Account number where the amount is settled	AlphaNumeric				ifsc_code	IFSC code of the account number	AlphaNumeric				pay_amount	Amount settled	Numeric				settlementID	Settlement id		32		Settlement ID for merchant or service provider	settlement_currency	Settlement Currency	Numeric	3		356	trans_currency	Settlement Currency	Numeric	3		356	utr_no	UTR number		24		
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utr_no	UTR number		24																																																		

secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix
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Settlement Details Request Parameters

This API needs to be called after settlement summary API for each settlementID received in the response. This API provides details of all the transactions that were settled under a particular settlementID. If there are more than 100 transactions settled under a particular settlementID then one API call will return only 100 transactions. Another API call with the same settlementID and passing the last transaction id needs to be made to get next set of 100 transactions and so on to get all transactions under that settlementID.

Parameter	Description	Type	Size	Mandatory	
merchantID	Merchant account ID	ANS	12	C	Either of merchantID OR aggregatorID should be present
aggregatorID	Aggregator ID	ANS	12	C	Either of merchantID OR aggregatorID should be present
providerID	Service provider ID	Alphanumeric		C	If the settlement id is for service provider
settlementID	SettlementID	ANS	32	Yes	
lastTxnID	Last txnid from the previous request.	Alphanumeric		No	
secureHash	Hash value	Alphanumeric	64	No	Check appendix for hash logic

This is a server to server API Call which uses POST method with content type as "application/json"

Settlement Details Response Parameters

Parameter	Description	Type	Size	Mandatory	
error_code	Response code indicating success/failure of the request	ANS	3	M	000 / 0000 – Success Any other value indicates status request failure / original transaction not found For status of the response always check txnStatus and txnResponseCode
merchantID	Merchant account ID			M	Echo of request field
lasttxnID	Last txnid in the response	Alphanumeric		No	
Payouts					
merchantID	Merchant account ID	Alphanumeric		C	Either merchantID or providerID will be available
providerID	Service Provider ID	Alphanumeric		C	Either merchantID or providerID will be available
merchantTxnNo	Merchant Txn ref Number	Alphanumeric			
settlementStatus	Settlement status of the transaction	String	3	C	NSD – Not yet settled STD – Settled
txnStatus	Status of the original transaction	String	3	C	REQ – Request received and in process SUC – Transaction Successful REJ – Transaction Rejected

					ERR – Error in transaction process
txnResponseCode	Txn response code	Alphanumeric			Original txn response code 000 / 0000 – Success Any other value indicates failure
txnID	Txn ID generated by PG	Alphanumeric	24	C	
paymentID	The authorization/ payment ID generated by the authorization system	Alphanumeric	24	C	
settlementAccount	Account number where the transaction was settled		21	M	
settlementAccountIFSC	IFSC code of the account number	Alphanumeric	8	M	
txnAmount	Txn amount	Numeric		M	
txnCharges	Total charges	Numeric		C	Only available in case of merchant settlement
serviceTax	Service tax paid	Numeric		C	Only available in case of merchant settlement
settledAmount	Amount settled in the account			M	Amount settled with merchant/service provider
utr_no	Utr number of the transfer	AlphaNumeric		M	

settlementDate	Date on which settlement was done	AlphaNumeric		M	YYYYMMDD
transmissionDateTime	Time at which transaction was executed	AlphaNumeric		M	YYYYMMDD
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Get Terminal Last Transaction

Parameter	Description	Type	Size	Mandatory	
terminalID	terminalID	ANS	12	Yes	
requestID	Unique request ID	ANS	12	Yes	
requestDateTime	Date time when the request was initiated	ANS	14	Yes	YYYYMMDDHHMISS The requestDateTime should be in UTC and the value cannot be older than 5 mins of current time.
lastTxnNo	Last transaction ref no	ANS	20	No	The last transaction reference No for which the terminal has already received the response. This API would return the recent successful transaction that has happened after the last transaction ref.

					If no transaction has happened after the last transaction ref no then error would be returned in response. If this field is omitted, then the recent successful transaction would be returned.
keyAlias	keyAlias	ANS	12	Yes	Terminal MasterKey Alias
secureHash	Hash value	Alphanumeric	64	Yes	Check appendix for hash logic

This is a server to server API Call which uses POST method with content type as "application/json"

Sample Request body.

```
{
  "terminalID": "000",
  "requestID": "Request successful",
  "requestDateTime" : "20101101121214"
  "txnStatus": "SUC",
  "txnResponseCode": "000",
  "txnRespDescription": "Transaction successful",
  "txnID": "T20394302393",
  "paymentDateTime": "20160803101010",
  "txnAuthID" : "28288",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```


Get Terminal Last Transaction Response

Parameter	Description	Type	Size	Mandatory	
responseCode	Response code indicating success/failure of the request	ANS	3	M	000 / 0000 – Success Any other value indicates status request failure / original transaction not found For status of the response always check txnStatus and txnResponseCode
respDescription	Description of failure		45	M	Ignore for success response.
merchantId	Merchant account ID			C	
aggregatorID	Merchant Aggregator ID			C	
merchantTxnNo	Merchant Txn ref Number			C	If available in response, this field should be stored by the terminal and sent in subsequent requests for fetching new responses.
txnStatus	Status of txn		3	C	REQ – Request received and in process SUC – Transaction Successful REJ – Transaction Rejected ERR – Error in transaction process
txnResponseCode	The response code for the original txn		24	C	Original txn response code 000 / 0000 – Success Any other value indicates failure

txnRespDescription	The unique ref number generated by the gateway		24	C	
txnID	The unique ref number generated by the gateway		24	C	
paymentDateTime	Authorization date time			C	YYYYMMDDHHMISS
txnAuthID	The authorization/payment ID generated by the authorization system		24	C	
secureHash	Hash value	Alphanumeric	64	Yes	Check Hash Calculation in Appendix

Sample response.

```
{
  "responseCode": "000",
  "respDescription": "Transaction successful",
  "merchantId": "123243",
  "merchantTxnNo" : "343223"
  "txnStatus": "SUC",
  "txnResponseCode": "000",
  "txnRespDescription": "",
  "txnID": "T20394302393",
  "paymentDateTime": "20160803101010",
  "txnAuthID" : "28288",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```

Sample HTML for Standard Mode Integration

Request Parameters HTTP- POST: https://PayPhi.co.in/Phiwebpay/v1.0/payments

```
<form id='myForm' name='myForm' action='${saleRequestURL}' method='post'
enctype="application/x-www-form-urlencoded">

    <input type="hidden" name="merchantID" value="M00001" />

    <input type="hidden" name="merchantTxnNo" value="M99887766" />

    <input type="hidden" name="amount" value="1000" />

    <input type="hidden" name="currencyCode" value="356" />

    <input type="hidden" name="payType" value="0" />

    <input type="hidden" name="customerEmailId" value="customer@email.com" />

    <input type="hidden" name="transactionType" value="SALE" />

    <input type="hidden" name="returnURL"
value="https://shoppingdomain.com/paymentResponseURL" />

    <input type="hidden" name="txnDate" value="20160803101010" />

    <input type="hidden" name="customerMobileNo" value="9123456789" />

    <input type="hidden" name="secureHash"
value="062f43817ff97eee69e89ff6d4ea283295db5d3f73a5c8addfa9012aed6aa995" />

</form>
```

Sample HTML for Direct Mode Integration

```
<form id='myForm' name='myForm' action='${saleRequestURL}' method='post'
enctype="application/x-www-form-urlencoded">

    <input type="hidden" name="merchantID" value="M00001" />

    <input type="hidden" name="merchantTxnNo" value="M99887766" />

    <input type="hidden" name="amount" value="1000" />

    <input type="hidden" name="currencyCode" value="356" />

    <input type="hidden" name="payType" value="1" />

    <input type="hidden" name="customerEmailId" value="customer@email.com" />

    <input type="hidden" name="transactionType" value="SALE" />

    <input type="hidden" name="returnURL"
value="https://shoppingdomain.com/paymentResponseURL" />

    <input type="hidden" name="txnDate" value="356" />

    <input type="hidden" name="customerMobileNo" value="9123456789" />

    <input type="hidden" name="cardNo" value="4111111111111111" />

    <input type="hidden" name="cardExp" value="202008" />

    <input type="hidden" name="nameOnCard" value="Customer Name" />

    <input type="hidden" name="cvv" value="123" />

    <input type="hidden" name="secureHash"
value="062f43817ff97eee69e89ff6d4ea283295db5d3f73a5c8addfa9012aed6aa995" />

</form>
```

Payment Response

```
<form id='myForm' name='myForm' action='${merchantReturnURL}' method='post'
enctype="application/x-www-form-urlencoded">

    <input type="hidden" name="responseCode" value="000" />

    <input type="hidden" name="respDescription" value="SUCCESS" />

    <input type="hidden" name="merchantId" value="M0000001" />

    <input type="hidden" name="merchantTxnNo" value="M0099999221" />

    <input type="hidden" name="txnID" value="T1472640294491" />

    <input type="hidden" name="paymentDateTime" value="20160831041454" />

    <input type="hidden" name="txnAuthID" value="006503" />

    <input type="hidden" name="secureHash"
value="39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2" />

</form>
```

Refund/Auth Response

Sample response for refund/auth would be in json format. Sample response given below.

```
{
  "responseCode": "000",
  "respDescription": "",
  "merchantId": "123243",
  "merchantTxnNo" : "343223"
  "txnID": "T20394302393",
  "paymentDateTime": "20160803101010",
  "txnAuthID" : "28288",
  "addlParam1" : "",
  "addlParam2" : "",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```

Transaction Status Response

Sample response for status would be in json format. Sample response given below.

```
{
  "responseCode": "000",
  "respDescription": "",
  "merchantId": "123243",
  "merchantTxnNo" : "343223"
  "txnStatus": "SUC",
  "txnResponseCode": "000",
  "txnRespDescription": "",
  "txnID": "T20394302393",
  "paymentDateTime": "20160803101010",
  "txnAuthID" : "28288",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```

Settlement Status Response

Sample response for settlement status would be in json format. Sample response given below.

```
{
  "responseCode": "000",
  "respDescription": "",
  "merchantId": "123243",
  "aggregatorID": "123243",
  "merchantTxnNo" : "343223"
  "settlementStatus": "STD",
  "txnStatus": "SUC",
  "txnResponseCode": "0000",
  "txnID": "T20394302393",
  "settlementID" : "28288_20191010-001",
  "settlementDate": "20160803",
  "secureHash": "39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2"
}
```

Hash Calculation

Step 1: Concatenate the parameter values (if not null and not an empty string) ***in ascending order of parameter names***. E.g., if param1 = "abc", param2 = "xyz" and name = "aa" then the concatenated value would be "aaabcxyz" (i.e., value of name + value of param1 + value of param2).

Step 2: Use HMAC function to calculate the hash. The algorithm used should be SHA256. The key used should be the key shared with the merchant/pg by PhiCommerce.

Step 3: Convert the result to HEX format.

Step 4: Convert the HEX format to lowercase and send it in secureHash parameter. E.g., if result of step 3 is "FEAB12CD" then convert it to "feab12cd". This step can be skipped if the result of step 3 is already in lowercase.

Note:

Don't ignore any parameters which is part of a response or request for hash calculation, even if the parameter is not part of published spec. Parameters which are not part of specs can be ignored for business processing but not for hash calculation. One can ignore only those parameters which are null or having empty values for hash calculation.

Java example for HMAC

```
public static String hmacDigest(String msg, String keyString) {
    String digest = null;
    try {
        SecretKeySpec key = new SecretKeySpec((keyString).getBytes("UTF-8"), "HmacSHA256");
        Mac mac = Mac.getInstance("HmacSHA256");
        mac.init(key);

        byte[] bytes = mac.doFinal(msg.getBytes("ASCII"));

        StringBuffer hash = new StringBuffer();
        for (int i = 0; i < bytes.length; i++) {
            String hex = Integer.toHexString(0xFF & bytes[i]);
            if (hex.length() == 1) {
                hash.append('0');
            }
            hash.append(hex);
        }
    }
}
```



```

        digest = hash.toString();
    } catch (UnsupportedEncodingException e) {
    } catch (InvalidKeyException e) {
    } catch (NoSuchAlgorithmException e) {
    }
    }
    return digest;
}

```

Java sample for server to server API call (HTTPS Post)

```

// HTTP POST request
public void sendPost() throws Exception {
    String url="https://abc.example.com/someoperation";
    URL obj=newURL(url);
    HttpURLConnection con =(HttpURLConnection)obj.openConnection();

    //add request header
    con.setRequestMethod("POST");
    con.setRequestProperty("Content-Type",
        "application/x-www-form-urlencoded");

    String
urlParams="merchantId=12321&merchantTxnNo=38928332&transactionType=REFUND&secureHash
=39867b8ef38a2853aacb84bbf5f927a72b27bdb301365245564f012d20dbc2f2";

    // Send post request
    con.setDoOutput(true);
    con.setDoInput(true);
    DataOutputStreamwr=newDataOutputStream(con.getOutputStream());
    wr.writeBytes(urlParams);
    wr.flush();
    wr.close();

    System.out.println("Sending POST request to URL : "+url);

    intresponseCode=con.getResponseCode();
    System.out.println("Post parameters : "+urlParameters);

    BufferedReader in =newBufferedReader(
        newInputStreamReader(con.getInputStream()));
    String inputLine;
    StringBuffer response =newStringBuffer();
    while((inputLine=in.readLine())!= null){
        response.append(inputLine);
    }
    in.close();

    //print result
    System.out.println(response.toString());
}

```

```
}
```

URLs

Test Environment URLs

Sale/PreAuth

<https://qa.phicommerce.com/pg/api/sale>

Refund/Auth/StatusCheck

<https://qa.phicommerce.com/pg/api/command>

Production URLs

Sale/PreAuth

<https://secure-ptg.payphi.com/pg/api/sale>

Refund/Void/Auth/StatusCheck

<https://secure-ptg.payphi.com/pg/api/command>

